

ALGEBRA QUALIFYING EXAMINATION

Fall 2024

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- (20 points) Part I is asking for statements of some definitions/theorems. Two questions, give concise but accurate answers.
- (64 points) Part II is True/False. Four questions, give a sketch proof or counterexample for full credit.
- (20+96 points) Part III consists of one medium-length problem worth 20 points then three longer problems worth a total of 96 points. Give concise but clear proofs stating clearly any results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following:

1. For a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, what it means for F to be an equivalence of categories.

Solution. There exists a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ such that $F \circ G$ is isomorphic to the identity functor of $\mathcal{D}$, and $G \circ F$ is isomorphic to the identity functor of $\mathcal{C}$.

2. For an R -module M , two different but equivalent definitions of what it means for M to be injective.

Solution. M is injective if $\text{Hom}(-, M)$ is exact. Equivalently, for any injective maps $P \rightarrow Q$ and any module map $P \rightarrow M$, there is a map $Q \rightarrow M$ making the triangle commute.

Part II. True or false? Provide a brief sketch of a proof (of truth or falsity), or provide a counterexample where relevant:

1. If U, V are finite-dimensional vector spaces and $f : U \rightarrow V, g : V \rightarrow U$ are linear maps then $\text{trace}(f \circ g) = \text{trace}(g \circ f)$.

Solution. Pick bases $u_1, \dots, u_m$ for U and $v_1, \dots, v_n$ for V , then let $A = [a_{ij}]_{1 \leq i \leq n, 1 \leq j \leq m}$ be the matrix of f and $[b_{ij}]_{1 \leq i \leq m, 1 \leq j \leq n}$. Then we have that

$$\text{trace}(f \circ g) = \text{trace}(AB) = \sum_{i=1}^n \sum_{j=1}^m a_{ij} b_{ji} = \sum_{j=1}^m \sum_{i=1}^n b_{ji} a_{ij} = \text{trace}(BA) = \text{trace}(g \circ f).$$

2. $\mathbb{Q}$ is a projective $\mathbb{Z}$ -module.

Solution. This is false. For every element $\alpha \in \mathbb{Q}^\times$ and every non-zero integer n there exists an element $\beta \in \mathbb{Q}^\times$ with $n\beta = \alpha$. This shows that $\mathbb{Q}$ cannot be a submodule of a free $\mathbb{Z}$ -module, and so, it is not a direct summand.

3. Let M be a finitely generated module over a PID R . If $M/IM = \{0\}$ for all maximal ideals I of R then $M = \{0\}$.

Solution. This is true. We can decompose a finitely generated module M as a finite direct sum of finitely generated indecomposable modules $M_1 \oplus \cdots \oplus M_n$. If $M/IM = \{0\}$ then $M_i/IM_i = \{0\}$ for each i . By the classification, for each i , we either have that $M_i \cong R$ or $M_i \cong R/(p^n)$ for some irreducible p and $n \geq 1$. In the former case, let I be any maximal ideal; then $M_i/IM_i = \{0\}$ implies that $R/I = \{0\}$ which is a contradiction. In the latter case, take $I := (p)$, a maximal ideal; then $M_i/IM_i \cong R/(p) = \{0\}$, again a contradiction.

4. There is a ring isomorphism $\mathbb{C}[x, y]/(x^2 - y^2) \cong \mathbb{C}[x, y]/(x - y) \times \mathbb{C}[x, y]/(x + y)$.

Solution. This is false... it looks like the Chinese Remainder theorem, but these ideals are not comaximal. Said another way, the vanishing sets of $x + y$ and $x - y$ intersect, so their union is not a disjoint union. I would probably give 85% partial credit to someone who argued that the homomorphism sending $x \mapsto (x, x)$ and $y \mapsto (y, y)$ is not an isomorphism, though this is technically not sufficient.

One method is to use algebraic geometry, because the corresponding algebraic spaces cannot be isomorphic; one has one connected component (because we can identify its vanishing locus) and the other has two (because it is a product of rings).

Another option is to explicitly verify that $\mathbb{C}[x, y]/(x^2 - y^2)$ does not have any nontrivial idempotents. By the monic polynomial lemma that any element of $\mathbb{C}[x, y]/(x^2 - y^2)$ can be uniquely expressed as $f(y) + xg(y)$ for $f, g \in \mathbb{C}[y]$. Squaring this element one has $f(y)^2 + y^2g(y)^2 + 2xf(y)g(y)$. So for an idempotent one has $2f(y)g(y) = g(y)$. Since $\mathbb{C}[y]$ is a domain, either $g(y) = 0$, or $g(y) \neq 0$ and $2f(y) = 1$. In the former case $f(y)^2 = f(y)$ and we obtain only the trivial idempotents 0 and 1. In the latter case $f(y)$ is a constant but is equal to $f(y)^2 + y^2g(y)^2$ which is only a constant when $g(y) = 0$, a contradiction.

A third option is to argue that the groups of units are different ($\mathbb{C}^\times$ versus $\mathbb{C}^\times \times \mathbb{C}^\times$). To compute the units in $\mathbb{C}[x, y]/(x^2 - y^2)$ one can again use that any element is expressible as $f(y) + xg(y)$ and calculate, similarly to the above. If $(f_1(y) + xg_1(y))(f_2(y) + xg_2(y)) = 1$, then $g_1f_2 + g_2f_1 = 0$ and $f_1f_2 + y^2g_1g_2 = 1$. For reasons of degree in $\mathbb{C}[y]$ we need $g_1g_2 = 0$ and $f_1f_2 = 1$, so f_1 and f_2 are both constant functions. By the first equation this implies that g_1 and g_2 are colinear, so both are zero.

Part III. Give complete answers to the following. In the longer questions 2–4, parts marked with a (*) indicate that they rely on the successful completion of an earlier part of the problem; parts without a (*) can be accomplished even without succeeding on the previous parts, though they may rely on assertions made in earlier parts.

1. Let R be a ring. Prove that R is left semisimple if and only if every left ideal of R is generated by an idempotent.

Solution. Suppose R is left semisimple. Let I be a left ideal of R . By assumption, $R = I \oplus J$ for some left ideal J . Then $1 = e + f$ for $e \in I$ and $f \in J$. So $Re \subseteq I$. Conversely, if $x \in I$, then $x = x \cdot 1 = xe + xf$, which implies $x = xe$ (and $xf = 0$). It follows that e is an idempotent and $I = Re$.

Conversely, suppose that every left ideal is generated by an idempotent. Let $I = Re$ be a left ideal. It suffices to prove that ${}_R R = I \oplus J$ where $J = R(1 - e)$. Indeed, each $r \in R$ can be written as $r = re + r(1 - e)$, whence ${}_R R = I + J$. To see that the sum is direct, let $r \in Re \cap R(1 - e)$. Then $r = xe = y(1 - e)$ implies $r = xe = xee = y(1 - e)e = 0$.

2. This problem is about groups of size 21.

- (i). Prove that any group G of size 21 is isomorphic to a semidirect product $C_3 \rtimes_{\phi} C_7$.
- (ii). Classify all groups of size 21 up to isomorphism.
- (iii). For the rest of this problem, choose a non-abelian group G of size 21. Write down a presentation of G .
- (iv). Compute the sizes of all conjugacy classes in G . Hint: this can be done by hand, but you can save a lot of time with numerology instead! Note that G has a normal subgroup.
- (v). Compute the size of the abelianization $G/[G, G]$. Hint: again, you can do this by hand, but you do not need to!
- (vi). (*) Compute the dimensions of all the irreducible representations of G .

Solution.

- (i). By Sylow theory, G has a subgroup P of size 3 and a subgroup Q of size 7. Since 3 and 7 are prime we have $P \cong C_3$ and $Q \cong C_7$.

There is a theorem: if $P \cap Q = \{1\}$ and $P \cdot Q = G$ and Q is normal, then $G \cong P \rtimes Q$. The subgroup $P \cdot Q$ has size dividing 21 but divisible by both 3 and 7, so $G = P \cdot Q$. The subgroup $P \cap Q$ has size dividing both 3 and 7, so $P \cap Q = \{1\}$. It remains to show that Q is normal.

By Sylow theory, the number of Sylow 7-subgroups divides 3 and is equal to 1 modulo 7. Thus it is 1, and this Sylow subgroup must be normal.

- (ii). A semidirect product $P \ltimes Q$ is determined first by a homomorphism $\phi: P \rightarrow \text{Aut}(Q)$. Recall that $\text{Aut}(\mathbb{Z}/N\mathbb{Z}) = (\mathbb{Z}/N\mathbb{Z})^\times$, which is cyclic when N is prime. Thus $\text{Aut}(C_7) = C_6$. There are three homomorphisms $C_3 \rightarrow C_6$, sending the generator to either the identity or to one of the two elements of order 3. The latter two homomorphisms are intertwined by precomposition with an automorphism of C_3 , an operation which yields isomorphic semidirect products. Thus there are at most two isomorphism classes of groups $C_3 \ltimes C_7$. One is abelian $C_3 \times C_7$ from the trivial homomorphism, and one is not, so there are exactly two.
- (iii). An element of C_6 of order 3 is 2. The corresponding element of $\text{Aut}(C_7)$ sends $g \mapsto g^2$ for all $g \in C_7$. This is the conjugation action of the generator of C_3 , so we have

$$G \cong \langle r, s \mid r^3 = s^7 = 1, rsr^{-1} = s^2 \rangle.$$

- (iv). Because conjugacy classes are orbits, they have size dividing the size of G , so they have sizes in the set $\{1, 3, 7, 21\}$. The normal subgroup $Q = C_7$ has size 7 and contains the identity, a conjugacy class of size 1. This is only possible if Q is a union of classes of sizes $(1, 3, 3)$. The union of all conjugacy classes of size 1 form the center $Z(G)$. But $Z(G) = \{1\}$ which can be verified in several ways (from the presentation; Q is not central and is the only subgroup of size 7, a subgroup of size 3 is conjugate to P so if it is central then it is P , but P is not central; if $G/Z(G)$ is cyclic then G is cyclic).

The remainder of G has size 14 and no classes of size 1, so it must split as two classes of size 7.

Thus the classes have size $(1, 3, 3, 7, 7)$.

- (v). As a semidirect product $C_3 \ltimes C_7$, G admits a quotient map to C_3 , which is abelian. Thus $G/[G, G]$ admits a quotient map to C_3 . The size of $G/[G, G]$ must therefore be either 3 or 21, but it cannot be 21 since G is not abelian. Thus $G/[G, G] \cong C_3$.
- (vi). The number of one-dimensional representations is the size of C_3 . The number of total representations is the number of conjugacy classes. So the sizes of representations are $(1, 1, 1, a, b)$ where a and b divide 21, and $1^2 + 1^2 + 1^2 + a^2 + b^2 = 21$. The only option is $a = b = 3$.

3. We provide an incomplete character table for a mysterious group G .

C order	id	C_1	C_2	C_3	C_4	C_5	C_6
$\mathbb{1}$	1	1	1	1	1	1	1
L_1	1	1	1	1	-1	-1	-1
L_2	1	1	-1	1	1	-1	-1
L_3	1	1	-1	1	-1	1	1
X	2	2	0	-2	0	0	0
Y_1	2	-2	0	0	0	α	β
Y_2	2	-2	0	0	0	γ	δ

- (i). Compute the kernel of the representation X (i.e. the subgroup of elements which act on X trivially).
- (ii). Is $\{id\} \cup C_1 \cup C_3 \cup C_4$ a normal subgroup? Justify your answer.
- (iii). Is $\{id\} \cup C_1 \cup C_3$ a normal subgroup? Justify your answer.
- (iv). Decompose $X \otimes X$ into irreducible representations. Hint: You can guess the answer without computing many pairings, if you look at the column of C_3 .
- (v). Which irreducible representation is L_2^* ?
- (vi). I claim that $Y_1^* \cong Y_2$. Compute the values of α , β , γ , and δ . Note that you can not distinguish between Y_1 and Y_2 with the information given, so you can choose which is which.

Solution.

- (i). The kernel of X is the union of conjugacy classes which agree with the identity conjugacy class. In this case it is $\{id\} \cup C_1$.
- (ii). This is a normal subgroup, as it is the kernel of L_2 .
- (iii). This is a normal subgroup, as it is the intersection of the kernel of L_2 with the kernel of L_1 . It is also the kernel of $X \otimes X$.
- (iv). The character of $X \otimes X$ is $(4, 4, 0, 4, 0, 0, 0)$. The only way to get 4 in column C_3 using representations of total dimension 4 is to add together four one-dimensional representations. It is easy to verify that the only way to make the signs cancel and get the desired zeroes is if

$$X \otimes X \cong \mathbb{1} \oplus L_1 \oplus L_2 \oplus L_3.$$

- (v). Since L_2 has real character entries, $L_2^* \cong L_2$.

(vi). Now we finally need to do some work. When pairing row Y_1 against row 1 we deduce $2\alpha + 2\beta = 0$, or $\beta = -\alpha$. Similarly, $\delta = -\gamma$. When pairing column 1 against column C_5 we get $2\alpha + 2\gamma = 0$, so $\alpha = \delta = -\gamma = -\beta$. Let's rewrite everything in terms of α . When pairing Y_1 against Y_2 we get $4 + 4 - 2\alpha\bar{\alpha} - 2\alpha\bar{\alpha} = 0$, so $\alpha\bar{\alpha} = 2$. Thus α is a complex number of norm $\sqrt{2}$. Pairing Y_1 against itself we get $4 + 4 + 2\alpha\bar{\alpha} + 2\alpha\bar{\alpha} = 16$, which gives nothing new. In fact, no pairings will give additional information beyond this point.

However, the claim that $Y_1^* \cong Y_2$ implies that $\gamma = \bar{\alpha}$ and $\delta = \bar{\beta}$. So $\alpha = -\bar{\alpha}$ is a purely imaginary complex number. Thus $\alpha = \pm\sqrt{2}i$.

4. Let $\alpha : \mathbb{C}[s, t, u] \rightarrow \mathbb{C}[x, y, z]$ be the ring homomorphism $s \mapsto xy$, $t \mapsto xz$, $u \mapsto yz$.

- (i). Let ϕ be the corresponding morphism between algebraic sets. Describe ϕ explicitly.
- (ii). (*) Note that $\phi(1, 2, 3) = (2, 3, 6)$. Describe the preimage $\phi^{-1}(2, 3, 6)$. Hint: if you know the values of xy and xyz then you know the value of z .
- (iii). (*) Describe the preimage $\phi^{-1}(0, 1, 2)$.
- (iv). (*) Consider the preimage $\phi^{-1}(0, 0, 0)$. Describe the decomposition of this preimage into irreducible components, and indicate the prime ideal corresponding to each component.
- (v). The map α is injective, so it embeds $\mathbb{C}[s, t, u]$ into $\mathbb{C}[x, y, z]$. Prove that this is *not* an integral extension.
- (vi). Prove that there is a $\mathbb{C}$ -algebra homomorphism $\beta : \mathbb{C}[s, s^{-1}, t, t^{-1}, u, u^{-1}] \rightarrow \mathbb{C}[x, x^{-1}, y, y^{-1}, z, z^{-1}]$ induced by α .
- (vii). The map β is injective, so it embeds

$$\mathbb{C}[s, s^{-1}, t, t^{-1}, u, u^{-1}] \quad \text{into} \quad \mathbb{C}[x, x^{-1}, y, y^{-1}, z, z^{-1}].$$

Prove that this is an integral extension.

Solution.

- (i). $\phi : \mathbb{A}^3 \rightarrow \mathbb{A}^3$, sending $(a, b, c) \mapsto (ab, ac, bc)$.
- (ii). If $ab = 2$ and $ac = 3$ and $bc = 6$ then $(abc)^2 = 2 \cdot 3 \cdot 6 = 36$, so $abc = \pm 6$. If $abc = 6$ then $a = abc/bc = 1$, and $b = 2$ and $c = 3$ similarly. If $abc = -6$ then $a = -1$ and $b = -2$ and $c = -3$. Thus the preimage consists of two points $(1, 2, 3)$ and $(-1, -2, -3)$.
- (iii). If $ab = 0$ then either $a = 0$ or $b = 0$, whence either $ac = 0$ or $bc = 0$. Thus the preimage is empty.

- (iv). In the preimage we need at least two out of the three scalars a, b, c to be zero. So the preimage is $(a, 0, 0) \cup (0, b, 0) \cup (0, 0, c)$. This is the union of the vanishing sets of the prime ideals (x, y) , (x, z) , and (y, z) .
- (v). α is not an integral extension, because ϕ is not surjective, so the Lying Over theorem fails.
- (vi). Let $S \subset \mathbb{C}[s, t, u]$ be the multiplicative set generated by $\{s, t, u\}$ and let $T \subset \mathbb{C}[x, y, z]$ be the multiplicative set generated by $\{x, y, z\}$. The composition of α with the inclusion map $\mathbb{C}[x, y, z] \rightarrow T^{-1}\mathbb{C}[x, y, z]$ gives a homomorphism $\mathbb{C}[s, t, u] \rightarrow T^{-1}\mathbb{C}[x, y, z]$. The generating elements of S are sent to units, so by the universal property of the localization, there is an induced map $S^{-1}\mathbb{C}[s, t, u] \rightarrow T^{-1}\mathbb{C}[x, y, z]$.
- (vii). β is an integral extension. We need only prove that the generators of $T^{-1}\mathbb{C}[x, y, z]$ are integral over the image of β . For example, x is integral because it satisfies the polynomial equation $x^2 - st/u = 0$. Similarly, x^{-1} is integral because it satisfies the polynomial equation $(x^{-1})^2 - u/st = 0$. Analogous formulas apply for the other generators.